Computer Science & IT



Database Management System

File organization & Indexing

Lecture No. 03

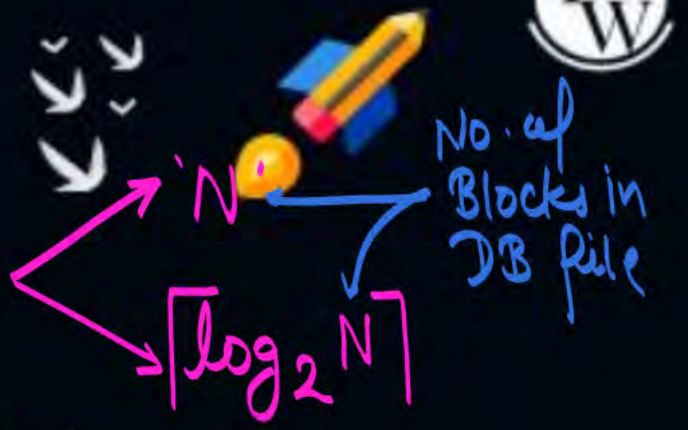


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Recap of Previous Lecture



- **Topic** IO cost without index file



- **Topic** Index file : Reduces IO cost
- **Topic** IO cost with indexing : $\lceil \log_2 M \rceil + 1$

M : No. of Blocks in index file

- **Topic** Categories of index
  - Dense Index { for each record of DB file, one entry in index file }
  - Sparse Index { for a set of DB record one entry in index file }

Slide

Topics to be Covered



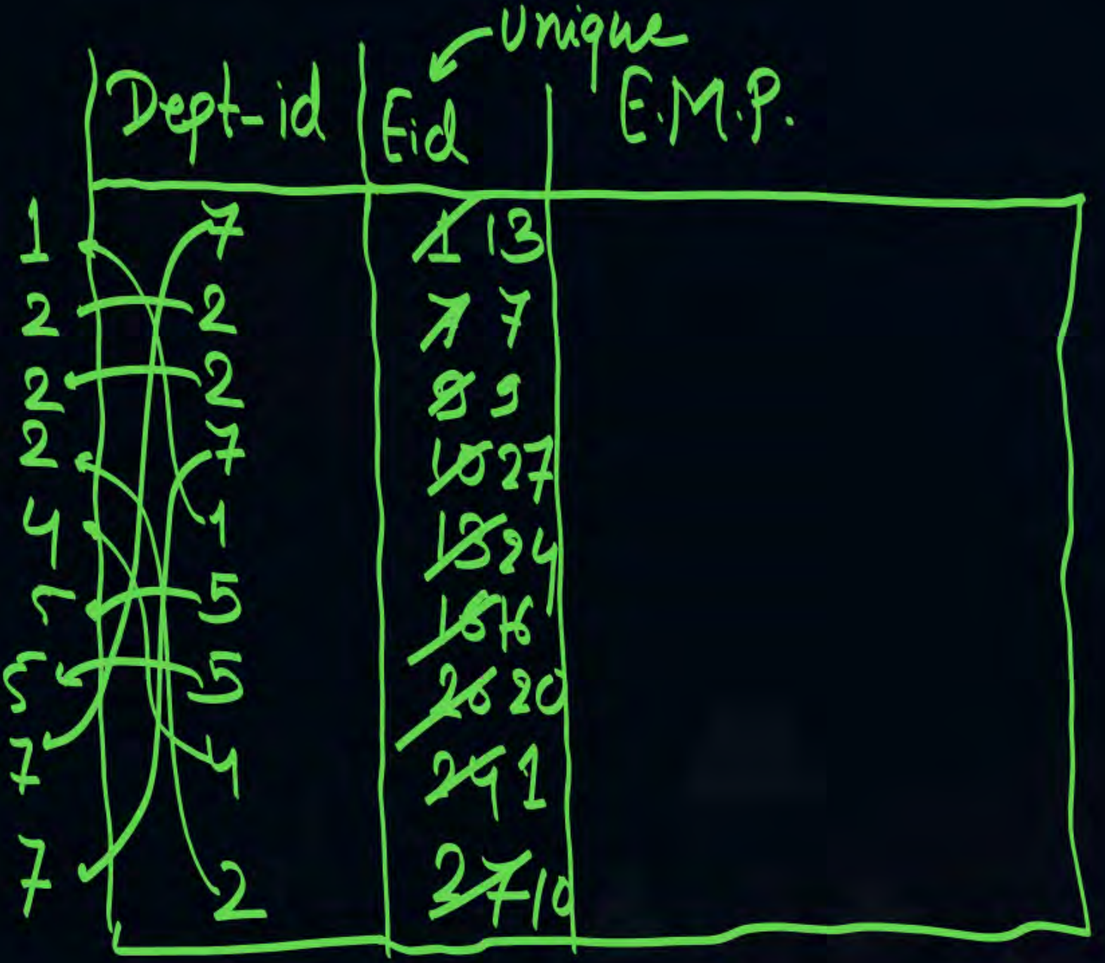
- Topic: Types of index
- Topic: Primary Index
- Topic: Clustering Index
- Topic: Secondary Index
- Topic: Multilevel Index
- Topic: B Tree Index

Single level index

Slide



Topic : Primary Index





Topic : Primary Index

If,

① Search key is a key attribute (unique values)

Primary key or Alternate key

and ② File is an ordered file based on search key

then, Index file created on such search key is called primary index.

Note: ① Primary index may be sparse index or dense index.

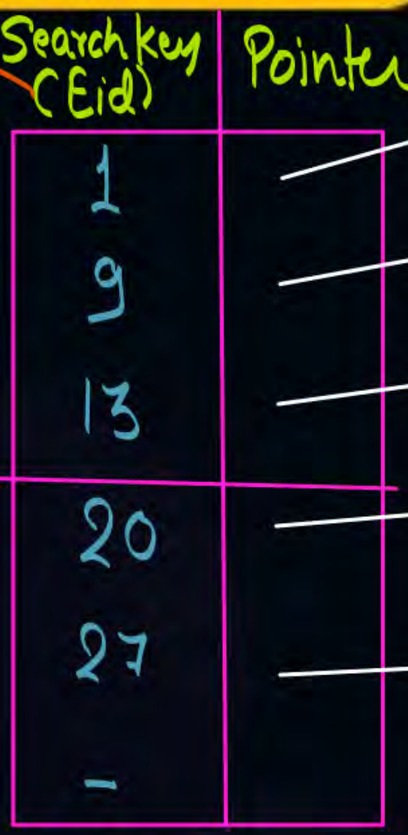
② At most one primary index is possible
{Because file must be ordered}



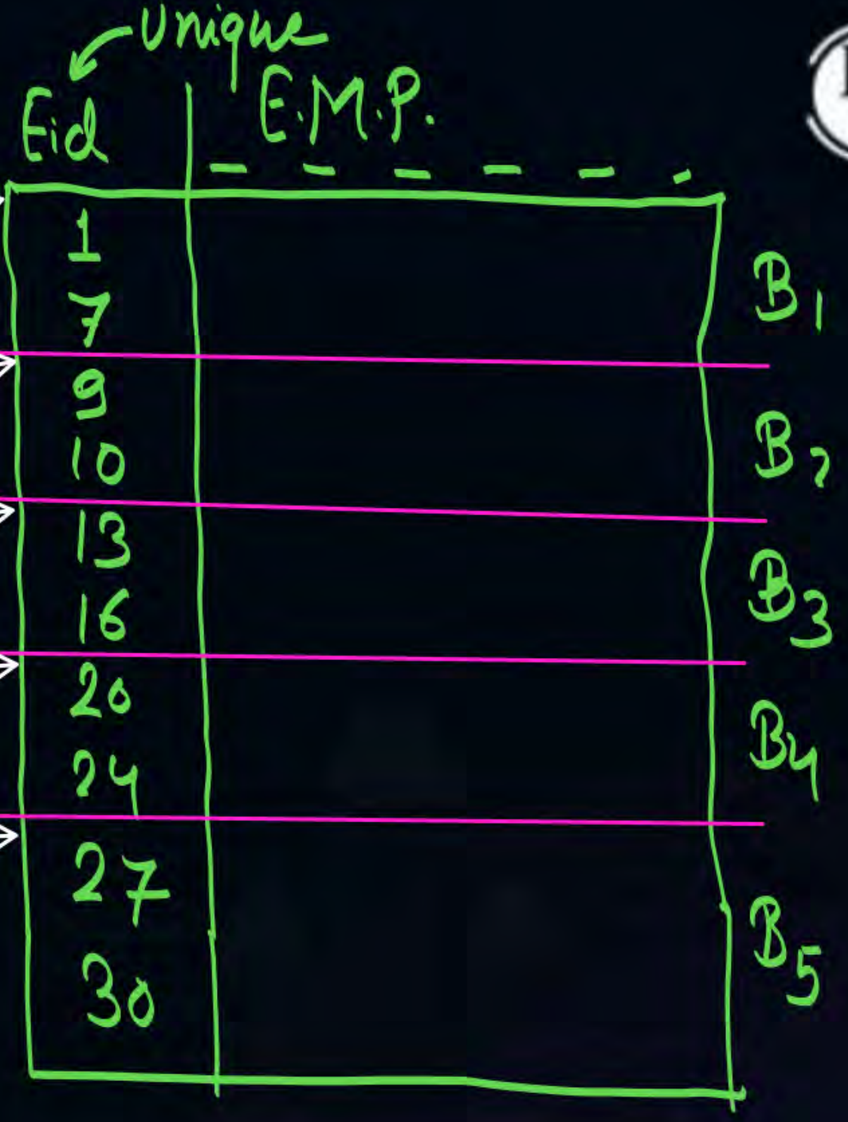
Topic : Primary Index

- Eid is a key attribute in EMP table,
  and
- Records of the EMP table are ordered based on Eid

∴ Index is a Primary index.



Index file
(Primary index)
{ It may be Sparse or dense }





Topic : Clustering Index

If:

① Search key is a non-key attribute. (values need not be unique.)

and ② File is ordered based on search key

then, Index file created on such search key is called clustering index

Note: ① Clustering index will always be sparse index

② At most one clustering index is possible



Topic : Clustering Index

- Dept_id is a non-key attribute in EMP table
  &
- Records of the EMP table are physically ordered based on Dept_id

∴ Index is Clustering index

Search key Dept_id | Pointer

Search key Dept_id	Pointer
1	
2	
4	
5	
7	

Index file (Clustering)

{Clustering index will always be Sparse index}

EMP

Dept_id	Eid	- - - -
1	7	
2	13	
2	1	
2	5	
4	4	
5	9	
5	15	
7	6	
7	20	
7	12	

B_1 B_2 B_3 B_4 B_5

Block Anchor



Topic : Clustering Index



- Dept_id is a non-key attribute in EMP table
- Records of the EMP table are physically ordered based on Dept_id

∴ Index is Clustering index

Search key Dept_id	Pointer
1	
2	

This is addr of the first record with search key = 2

If the record is the last record of the block, then we don't know whether it is the last record w.r.t. search key = 2 or not.

So we will use block anchor and we will access the next block of the file to check if there are more records w.r.t. search key = 2.

EMP

Dept_id	Eid	- - - -
1	7	
2	13	
2	1	
2	5	
4	4	
5	9	
5	15	
7	6	
7	20	
7	12	

B_1 B_2 B_3 B_4 B_5

Block Anchor

Note:- On a database file we will either have Primary index or clustering index but not both.

{ Because file must be ordered based on search key }



Topic : Secondary Index

* If,
  ① File is a unordered file based on search key then index file created on such search key is called secondary index.

{Note: Search key may be a Key attribute (or) Non-key attribute}
- Key attribute → May be unique
- or
- Non-key attribute → May not be unique

Note: ① Secondary index will always be dense index
② Any number of secondary index are possible
{Because file need not be ordered file}

Topic : Secondary Index



file is unordered based on Pass-No.
∴ Secondary index

Pass-No.	Pointer
1, 2, 4, 7	
9, 10, 12, 13	

file is unordered based on rating
∴ Secondary index

Rating	Pointer

Flyer database

Pass-No.		Rating
4, 1		1, 2
9, 2		3, 2
13, 7		5, 3
10, 12		4, 5

Topic : Secondary Index



file is unordered based on Pass-No.
∴ Secondary index

Pass-No.	Pointer
1	
2	
4	
7	
9	
10	
12	
13	

file is unordered based on rating
∴ Secondary index

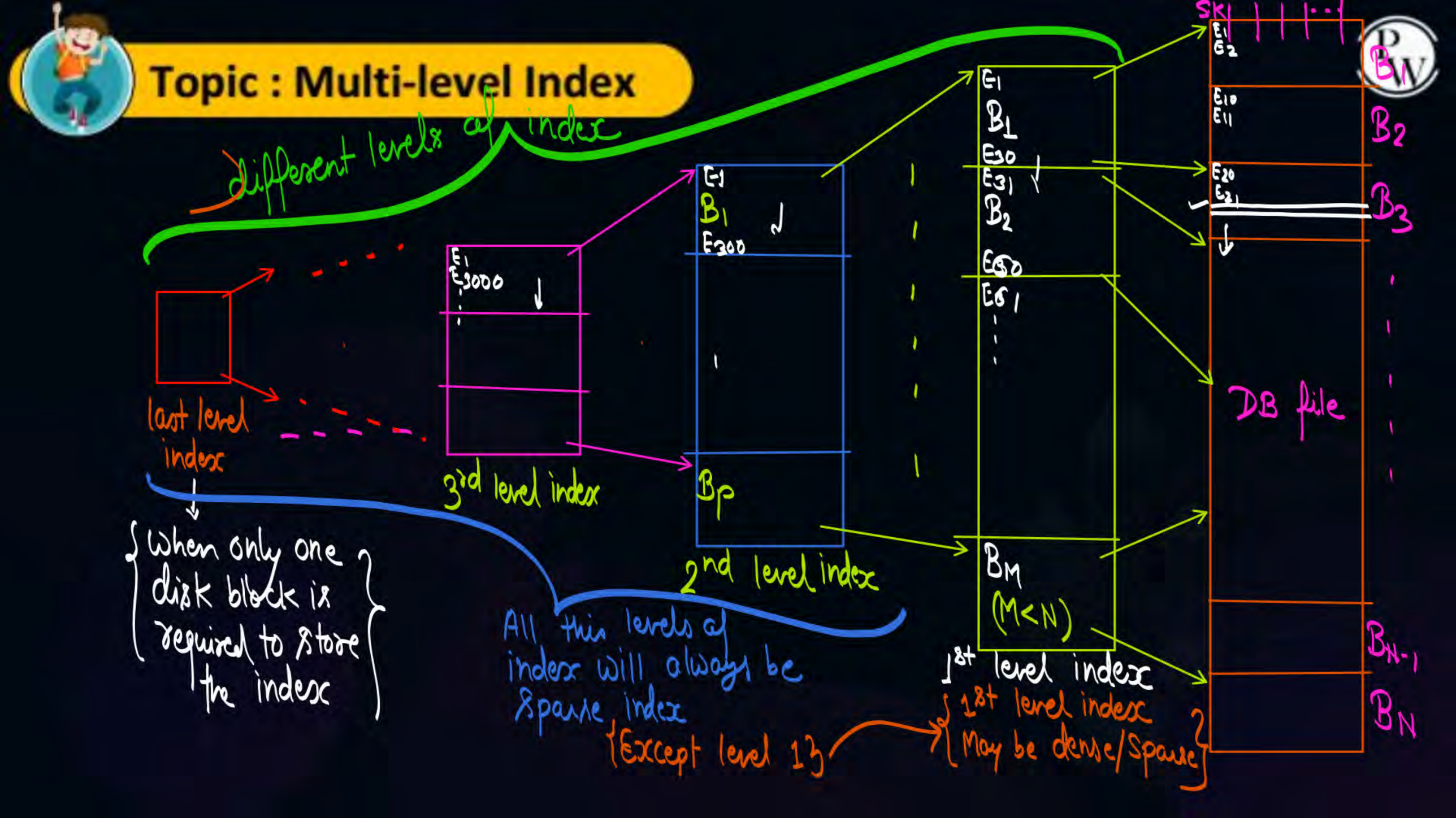
Rating	Pointer
1	
2	
2	
3	
3	
4	
5	
5	

Flyer database

Pass-No.		Rating
4		1
1		2
9		3
2		2
13		5
7		3
10		4
12		5

Note: If file is unordered based on search key then 1st level index file created on such search key will always be dense index

Because the first task will be order the index file base on search key values.


Topic : Multi-level Index
different levels of index
last level index
3rd level index
2nd level index
1st level index
DB file
When only one disk block is required to store the index
All this levels of index will always be Sparse index
{Except level 1}
{1st level index May be dense/Sparse}



Topic : Multi-level Index

Note: If index is a multi-level index, and there are 'S' levels of index, then

IO Cost of an access is = S' + 1
(with multi-level index) = No. of levels + 1

from each level of index one block will be accessed

To access the block of DB file based on the addr provided by inner most level of index



Topic : Multi-level Index

i.e. the database in which insertion and deletion are very rare

- Above multi-level index is suitable only for "static database", because if we insert a new record in the DB file or if we delete an existing record from the DB file, then we may have to restructure of all the levels of index, and it will be very time consuming.
  - If database is dynamic database {i.e. insertion and/or deletion are frequent}, then most of the time will be wasted in maintaining the index

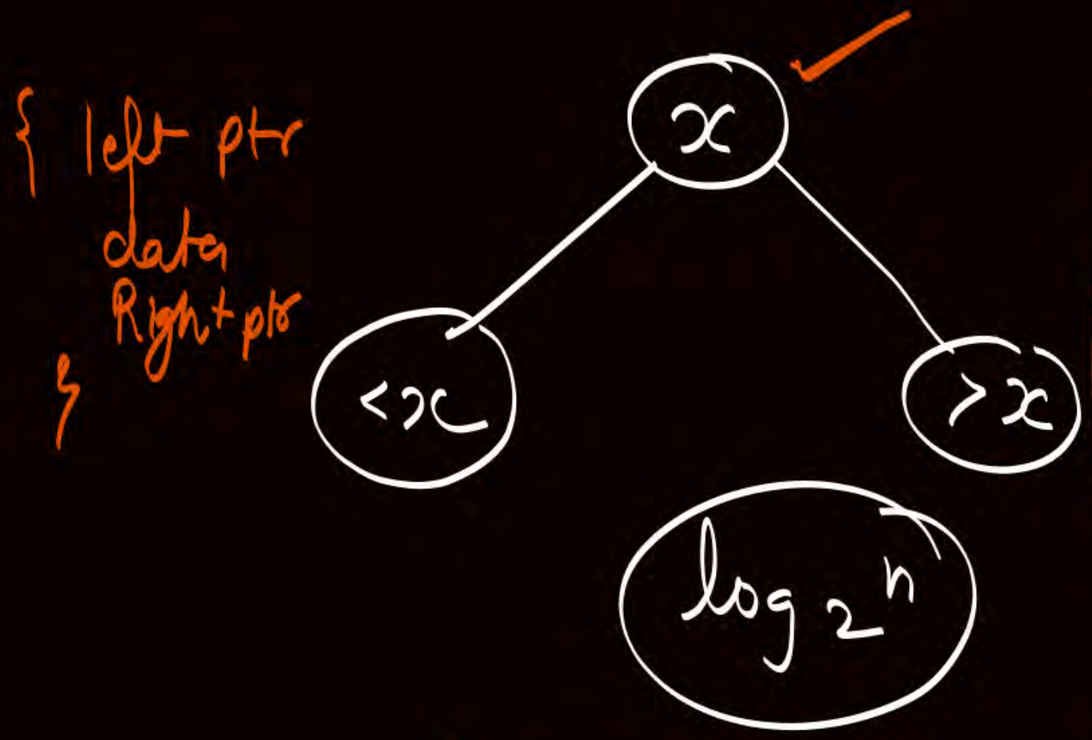


Topic : Dynamic Multi-level Index

- For the database in which insertion and/or deletion are very frequent we must use dynamic multi-level index.

- Dynamic multilevel index are self balancing index trees.

eg: → B Tree
→ B+ Tree



2-Way Search tree

{i.e Binary S.T}



{ P_1 data 1 P_2 data 2 P_3 : : data n-1 P_m }

{ m-ary Search tree or M-way Search tree }



2 mins Summary

- Topic: Types of index
- Topic: Primary Index
- Topic: Clustering Index
- Topic: Secondary Index
- Topic: Multilevel Index
- Topic: B Tree Index

Slide

THANK - YOU

Slide